

6 Numerical experiments

We first collect the convergence and unforced energy-dissipation experiments for both time discretizations, and then present two magnetically forced channel-flow applications of the second-order scheme.

6.1 Experimental setup

We assess both time discretizations by a manufactured-solution test on $\Omega = (0, 1)^3$ with $T = 2$. Each Cartesian partition into $K \times K \times K$ cubes is subdivided into tetrahedra, with $K = 4, 8, 16, 32$ and $\Delta t = 1/K = h/\sqrt{3}$, where h is the maximum tetrahedron diameter. Space and time are refined together; this experiment does not isolate the temporal convergence order.

Manufactured solution. Define

$$q(s) = 64s^3(1-s)^3, \quad B(x, y, z) = q(x)q(y)q(z), \quad S(x, y, z) = \sin^2(\pi x) \sin^2(\pi y) \sin^2(\pi z),$$

and let $a = (1, 2, 3)^\top$, $b = (1, -1, 1/2)^\top$, and $c = (1, 2, -1)^\top$. The prescribed fields are

$$\begin{aligned} \mathbf{u} &= \frac{1 + 0.1t}{C_u} \nabla B \times a, & \boldsymbol{\omega} &= \frac{1 - 0.05t}{C_\omega} Bb, & \mathbf{m} &= \frac{1 + 0.05t}{C_m} Sc, \\ \varphi &= \frac{1 - 0.05t}{C_\varphi} B, & \mathbf{H} &= \nabla \varphi, & p &= \frac{1 + 0.05t}{C_p} \prod_{s \in \{x, y, z\}} \cos(2\pi s), \end{aligned}$$

where the fixed normalization constants used in the computations are

$$\begin{aligned} C_u &= 363.275188821, & C_\omega &= 20.6500992827, & C_m &= \frac{\sqrt{81 + 108\pi^2}}{16}, \\ C_\varphi &= 13.7667328551, & C_p &= 0.353553390593. \end{aligned}$$

The auxiliary fields are $\mathbf{k} = \text{curl } \mathbf{m}$, $\mathbf{z} = \mathbf{u} \times \mathbf{m}$, $\vartheta = -\Delta \varphi$, and $\varpi = \mathbf{u} \cdot \mathbf{m}$. We set $\mathbf{H}_e = -\mu_0(\mathbf{m} + \mathbf{H})$ and use the zero-mean modified pressure $\tilde{p} = \mathcal{P}_0[p - \mu_0 \mathbf{m} \cdot \mathbf{H}/2]$, where $\mathcal{P}_0 v = v - |\Omega|^{-1} \int_\Omega v$. All physical parameters $\rho, \kappa, \eta, \zeta, \mu_0, \sigma, \eta', \lambda', \tau, \chi_0$ are set to one.

This is a forced verification problem. The momentum, angular-momentum, and magnetization sources $\mathbf{f}, \mathbf{g}, \mathbf{r}$ are obtained by substituting the prescribed fields into the corresponding continuous equations. Their weak right-hand sides are $(\mathbf{f}, \mathbf{v}_h)$, $(\mathbf{g}, \mathbf{s}_h)$, and $(\mathbf{r}, \mathbf{F}_h)$; the magnetic-potential right-hand side is

$$\left(-\partial_t \mathbf{H}_e - \frac{\mu_0 + \chi_0}{\tau \mu_0} \mathbf{H}_e + \sigma \nabla \text{div } \mathbf{H}_e - \mu_0 \mathbf{r}, \nabla \psi_h \right).$$

The fields satisfy the homogeneous boundary traces used in the experiment: zero velocity and angular velocity, zero normal magnetization, zero tangential $\mathbf{z}, \mathbf{k}, \mathbf{H}$, and zero $\varphi, \vartheta, \varpi$. In particular, q, q', q'' vanish at both endpoints and $S, \nabla S$ vanish on the cube boundary. The new ϑ and ϖ couplings are not identically zero for this test.

Discretization and implementation. The first-order runs use the MINI velocity element, continuous piecewise linear pressure and angular velocity, lowest-order Raviart–Thomas magnetization and first-family Nedelec $\mathbf{z}, \mathbf{k}, \mathbf{H}$, and continuous piecewise linear $\varphi, \vartheta, \varpi$.

For the second-order scheme, velocity and angular velocity use continuous piecewise quadratic elements, pressure remains continuous piecewise linear, magnetization uses RT1, and $\mathbf{z}, \mathbf{k}, \mathbf{H}$ use first-family Nedelec elements one degree above the lowest order. The scalar fields $\varphi, \vartheta, \varpi$ are continuous piecewise quadratic. The corresponding RT and Nedelec Basix degree is two.

The initial velocity is obtained by a discrete Stokes projection. After interpolating the initial magnetization into its RT space, the initial potential is determined from the compatible magnetic Poisson problem. The initial $\mathbf{k}, \mathbf{z}, \vartheta, \varpi$ satisfy their discrete mass-projection relations. Pressure is normalized by its finite-element integral mean.

For the second-order convergence test, the first time step is solved by a Broyden quasi-Newton iteration with relative tolerance 10^{-9} and a maximum of 12 iterations. The accepted state is checked by reassembling the nonlinear midpoint equations. Subsequent time steps use the prescribed extrapolated coefficients and require one five-block linear solve per step. The $\mathbf{z}_h$ and ϖ_h slots represent the current midpoint-stage projections; they are not averaged again with the preceding stage.

Computations use FEniCSx/DOLFINx 0.10.0 [1] with PETSc and MUMPS. The five-block Algorithm 3 is evaluated through matrix-free Schur-complement actions. The K_1, K_3, K_5 subproblems use MUMPS LU; the S_2, S_4 systems use FGMRES with LU preconditioners. The MUMPS ordering control is ICNTL(7) = 7. The Schur tolerances are 10^{-10} relative and 10^{-12} absolute. All first-order cases and the second-order cases up to $K = 16$ use 48 MPI processes with one thread each; the second-order $K = 32$ case uses 32 MPI processes with four threads

each on a 1-TB-class node. Source terms are evaluated at quadrature points, with assembly quadrature degree 12. In the error calculations, the analytic values and derivatives are independently represented in a degree-6 DG space and integrated with degree-12 quadrature; the numerical solution is evaluated in its original finite-element space.

As implementation checks for the manufactured-solution runs, the two schemes were compared with a monolithic solve of the same matrix at the first two time levels on $K = 4$, modulo the same pressure nullspace; the normalized solution differences were below 1.1×10^{-12} . An independent transcription of the nine weak equations on $K = 2$, using random finite-element fields and both unit and non-unit parameters, also agreed with the production matrices and right-hand sides to scaled differences below 1.1×10^{-16} . These checks concern algebraic and implementation consistency, rather than replacing the convergence test.

Error measures. For a field v and its specified norm X , we report

$$E_X(v) = \frac{\|v - v_h\|_X}{\|v\|_X}, \quad \text{rate}(K \rightarrow 2K) = \log_2 \frac{E_X(v; K)}{E_X(v; 2K)}.$$

The norms are full H^1 for $\mathbf{u}, \boldsymbol{\omega}, \varphi$, L^2 for $\tilde{p}, \mathbf{k}$, $H(\text{div})$ for $\mathbf{m}$, and $H(\text{curl})$ for $\mathbf{H}$. Thus the graph norms combine the squared L^2 and derivative terms before taking the square root. Pressure errors use zero-mean numerical and exact fields. First-order errors are evaluated at T .

For the second-order scheme, the primary fields are compared at T . Its pressure slot represents the averaged modified pressure, so its reference is

$$\tilde{p}_{\text{ref}}^{N-1/2} = \frac{\tilde{p}(T) + \tilde{p}(T - \Delta t)}{2},$$

with the same zero-mean convention as above. This reference is not identified with either the endpoint pressure or the analytic pressure evaluated directly at the midpoint.

We also report C_H , the maximum of $|\text{curl } \mathbf{H}_h|$ over the degree-6 DG interpolation points, as a sampled estimate of $\|\text{curl } \mathbf{H}_h\|_{L^\infty}$. It is an absolute diagnostic, not a relative error or a rigorous continuous supremum.

6.2 Convergence verification

We report the joint space–time refinement results separately for the two schemes, using the common test and error measures defined above.

6.2.1 First-order scheme

Table 1 gives the errors and observed orders; Figure 1 shows the corresponding log–log error curves. On the finest refinement interval, the rates for $\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \mathbf{k}, \varphi, \mathbf{H}$ are 1.019, 0.994, 0.999, 0.997, 1.018, and 1.017, respectively. The modified pressure exhibits a rate of 2.065 in this test. The sampled curl diagnostic remains between 1.22×10^{-13} and 8.51×10^{-12} , consistent with the discrete gradient recovery of the magnetic field up to numerical error.

Table 1: Relative errors and observed rates for the first-order scheme, $T = 2$ and $\Delta t = 1/K$. C_H is the sampled absolute curl diagnostic.

(a) Velocity, pressure, angular velocity, and magnetization								
K	$\mathbf{u}: H^1$		$\tilde{p}: L^2$		$\boldsymbol{\omega}: H^1$		$\mathbf{m}: H(\text{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
4	9.6554×10^{-1}	–	5.6019×10^{-1}	–	6.4182×10^{-1}	–	3.9160×10^{-1}	–
8	4.3070×10^{-1}	1.165	1.0260×10^{-1}	2.449	3.5671×10^{-1}	0.847	1.9873×10^{-1}	0.979
16	1.9858×10^{-1}	1.117	2.1172×10^{-2}	2.277	1.8331×10^{-1}	0.960	9.9738×10^{-2}	0.995
32	9.8002×10^{-2}	1.019	5.0596×10^{-3}	2.065	9.2018×10^{-2}	0.994	4.9919×10^{-2}	0.999
(b) Auxiliary curl field, potential, magnetic field, and curl diagnostic								
K	$\mathbf{k}: L^2$		$\varphi: H^1$		$\mathbf{H}: H(\text{curl})$		C_H	
	Error	Rate	Error	Rate	Error	Rate		
4	5.1511×10^{-1}	–	6.5848×10^{-1}	–	6.6223×10^{-1}	–	1.2220×10^{-13}	
8	2.6813×10^{-1}	0.942	3.7678×10^{-1}	0.805	3.7965×10^{-1}	0.803	5.1946×10^{-13}	
16	1.3523×10^{-1}	0.988	1.8757×10^{-1}	1.006	1.8935×10^{-1}	1.004	2.0267×10^{-12}	
32	6.7750×10^{-2}	0.997	9.2620×10^{-2}	1.018	9.3572×10^{-2}	1.017	8.5067×10^{-12}	

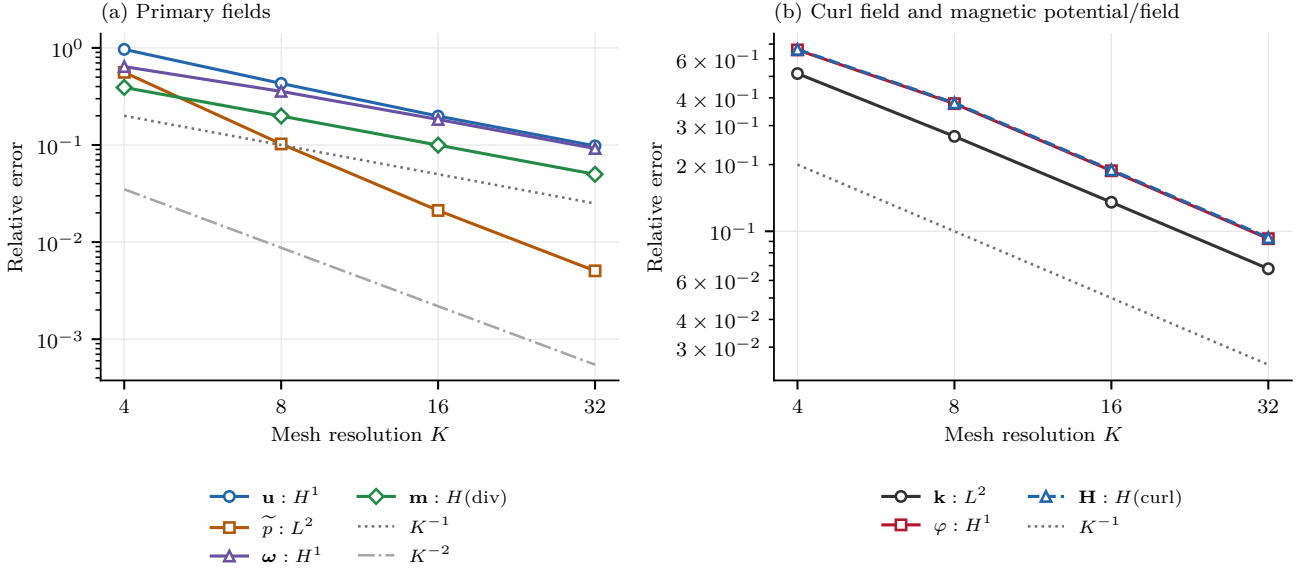


Figure 1: Relative errors for the first-order discretization under joint refinement with $\Delta t = 1/K$. (a) Velocity, modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is the mesh resolution $K = 4, 8, 16, 32$, so refinement proceeds from left to right. Reference lines proportional to K^{-1} and K^{-2} indicate first- and second-order decay with mesh refinement, not independently measured temporal orders.

6.2.2 Second-order scheme

Table 2 and Fig. 2 show that the finest-grid rates for $\mathbf{u}, \tilde{p}, \boldsymbol{\omega}, \mathbf{m}, \varphi, \mathbf{H}$ are 2.166, 2.051, 1.981, 2.007, 1.968, and 1.968, respectively. The L^2 error of $\mathbf{k}$ decreases approximately at first order, with a finest-grid rate of 1.015; second-order convergence is therefore not claimed for every auxiliary quantity. The sampled curl diagnostic ranges from 7.30×10^{-13} to 6.96×10^{-11} . For $K = 32$, all 64 steps reach $T = 2$, and the final full-system relative algebraic residual is 2.08×10^{-13} . These observations support the reported joint-refinement behavior. Because the mesh and time step vary together and the primary exact fields have linear temporal amplitudes, they do not constitute a separate test of pure temporal accuracy or of energy stability.

Table 2: Relative errors and observed rates for the second-order scheme, $T = 2$ and $\Delta t = 1/K$. C_H is the sampled absolute curl diagnostic.

(a) Velocity, pressure, angular velocity, and magnetization								
K	$\mathbf{u}: H^1$		$\tilde{p}: L^2$		$\boldsymbol{\omega}: H^1$		$\mathbf{m}: H(\text{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
4	5.6873×10^{-1}	—	4.2000×10^{-1}	—	2.1381×10^{-1}	—	1.1937×10^{-1}	—
8	1.5090×10^{-1}	1.914	9.4815×10^{-2}	2.147	6.2017×10^{-2}	1.786	3.1613×10^{-2}	1.917
16	2.4805×10^{-2}	2.605	2.0872×10^{-2}	2.184	1.6320×10^{-2}	1.926	7.8383×10^{-3}	2.012
32	5.5259×10^{-3}	2.166	5.0382×10^{-3}	2.051	4.1345×10^{-3}	1.981	1.9494×10^{-3}	2.007

(b) Auxiliary curl field, potential, magnetic field, and curl diagnostic								
K	$\mathbf{k}: L^2$		$\varphi: H^1$		$\mathbf{H}: H(\text{curl})$		C_H	
	Error	Rate	Error	Rate	Error	Rate		
4	2.1008×10^{-1}	—	2.0119×10^{-1}	—	2.0303×10^{-1}	—	7.3001×10^{-13}	
8	9.5804×10^{-2}	1.133	6.0098×10^{-2}	1.743	6.0718×10^{-2}	1.742	3.9064×10^{-12}	
16	4.6084×10^{-2}	1.056	1.6119×10^{-2}	1.899	1.6289×10^{-2}	1.898	1.8492×10^{-11}	
32	2.2804×10^{-2}	1.015	4.1189×10^{-3}	1.968	4.1627×10^{-3}	1.968	6.9611×10^{-11}	

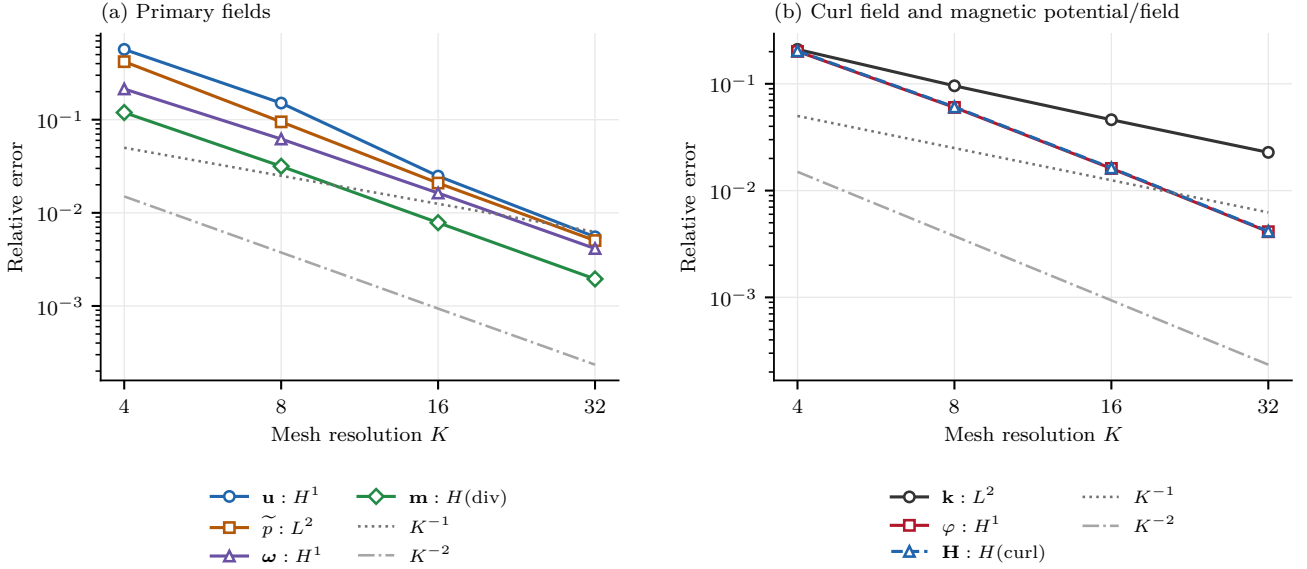


Figure 2: Relative errors for the second-order discretization using the same manufactured solution and $\Delta t = 1/K$. (a) Velocity, averaged modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is $K = 4, 8, 16, 32$, increasing from left to right. Reference lines proportional to K^{-1} and K^{-2} distinguish the nearly quadratic decay of the principal errors from the approximately first-order L^2 decay for $\mathbf{k}$.

6.3 Energy dissipation

We test free relaxation on the fixed $K = 8$ mesh, with $T = 0.5$ and $\Delta t = 1/16, 1/32, 1/64$. All physical parameters remain one. Unlike the manufactured-solution test, the applied field and all compensating sources are set to zero: $\mathbf{H}_e = \mathbf{f} = \mathbf{g} = \mathbf{r} = 0$. We retain the homogeneous boundary traces specified in Section 6.1. The initial shapes are

$$\mathbf{u}_0 = C_u^{-1} \nabla B \times \mathbf{a}, \quad \boldsymbol{\omega}_0 = C_\omega^{-1} B \mathbf{b}, \quad \mathbf{m}_0 = C_m^{-1} S \mathbf{c}.$$

The initial velocity is projected onto the discrete solenoidal space; angular velocity and magnetization are interpolated into their respective spaces. The initial potential is recomputed with zero boundary trace from

$$(\nabla \varphi_h^0, \nabla \psi_h) = -(\mathbf{m}_h^0, \nabla \psi_h) \quad \text{for all admissible } \psi_h,$$

rather than prescribed from the manufactured magnetic potential. The auxiliary variables satisfy their discrete projection relations. Thus the internal magnetic field is not artificially suppressed, and the subsequent solution is not constrained to follow the manufactured temporal profile.

At every time level we integrate the discrete energy

$$E_h^n = \frac{1}{2} (\rho \|\mathbf{u}_h^n\|^2 + \rho \kappa \|\boldsymbol{\omega}_h^n\|^2 + \|\mathbf{m}_h^n\|^2 + \mu_0 \|\nabla \varphi_h^n\|^2). \quad (1)$$

The norms in (1) are spatial L^2 norms. Energy and dissipation are evaluated in the original finite-element spaces using degree-12 quadrature, not from the visualization fields. For the first-order scheme, we check the balance defect

$$R_h^n = E_h^n - E_h^{n-1} + \Delta t D_h^n + N_h^n, \quad N_h^n = \frac{1}{2} \|\mathcal{X}_h^n - \mathcal{X}_h^{n-1}\|_{\mathbf{D}}^2, \quad (2)$$

where D_h^n is the discrete viscous, spin, magnetic-diffusion, and relaxation dissipation corresponding to $\mathbf{A}$. In particular, the weak magnetic diffusion terms are evaluated as $\sigma \|\mathbf{k}_h^n\|^2$ and $\sigma \mu_0 \|\vartheta_h^n\|^2$. For the second-order discretization, the balance defect is

$$R_h^n = E_h^n - E_h^{n-1} + \Delta t D_h^{n-1/2}, \quad (3)$$

where $D_h^{n-1/2}$ is evaluated at the midpoint averages of the primary fields and the endpoint auxiliaries $\mathbf{k}_h, \vartheta_h$. There is no first-order numerical-dissipation term N_h^n . The stage fields $\mathbf{z}_h, \varpi_h$ are not averaged a second time.

For both schemes, the reported balance diagnostic is

$$\epsilon_{\text{bal}}^n = \frac{|R_h^n|}{\max\{E_h^n, E_h^{n-1}, \Delta t D_h^n + N_h^n, 10^{-14} E_h^0\}},$$

with the midpoint dissipation and $N_h^n = 0$ used for the second-order scheme.

Both schemes retain the spatial approximation orders specified in Section 6.1. Thus their discrete initial states differ, whereas each scheme uses the same initial state for all three time steps.

All energy runs use 48 MPI processes with one thread each and the same five-block solver settings as the convergence experiments. The first-order runs require no nonlinear startup.

For the second-order energy test, the initial Broyden solve uses relative tolerance 10^{-10} with at most 20 iterations, and the accepted midpoint equations are independently reassembled to check their residual. Subsequent time steps use the prescribed extrapolation.

6.3.1 First-order scheme

Figure 3 shows monotone decay for all three time steps. The initial discrete energy is approximately 4.205865×10^{-2} ; the final energy ratios and balance diagnostics are given in Table 3. The largest balance diagnostic is 4.61×10^{-14} . These results verify dissipation for the tested discretizations; they do not constitute a pure temporal convergence study.

Table 3: First-order unforced energy test: $K = 8$, $T = 0.5$.

Δt	$E_h(T)/E_h(0)$	$\max_n \epsilon_{\text{bal}}^n$
1/16	9.1847×10^{-5}	3.69×10^{-14}
1/32	3.1955×10^{-5}	2.22×10^{-14}
1/64	1.6247×10^{-5}	4.61×10^{-14}

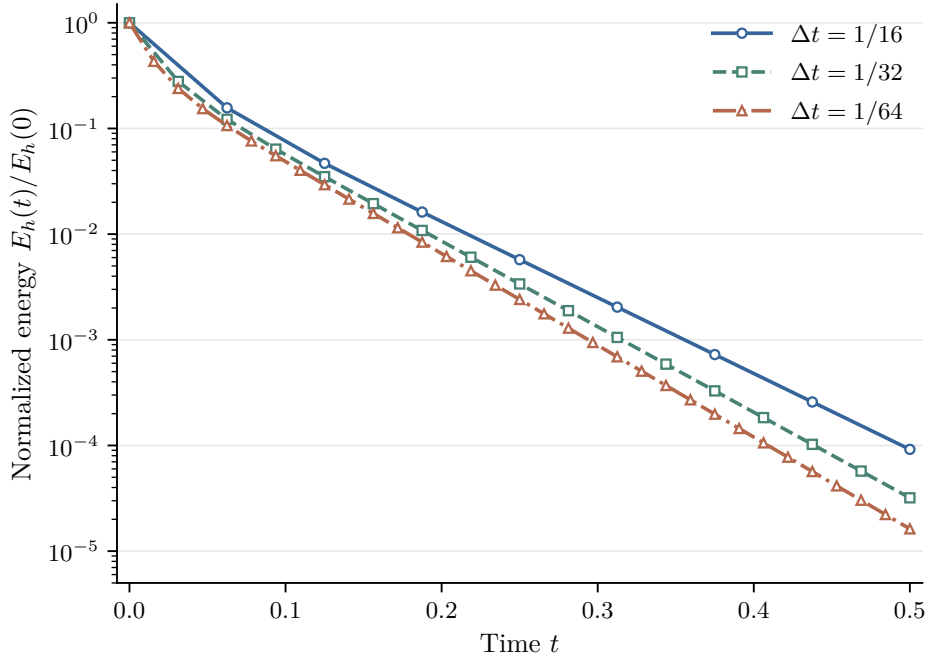


Figure 3: Normalized discrete energy for the first-order scheme in the unforced relaxation test, $K = 8$ and $T = 0.5$. The vertical axis is logarithmic. Symbols represent all computed time levels; connecting lines are not fitted or smoothed curves. The same discrete initial state is used for all three time steps.

6.3.2 Second-order scheme

Figure 4 shows monotone energy decay for all three time steps, with initial energy approximately 4.295033×10^{-2} . Table 4 reports the final ratios and balance diagnostics; the largest normalized balance defect is 3.71×10^{-11} . The coarsest time step produces slower late-time decay, which is retained in the plot. Energy dissipation therefore should not be interpreted as evidence that the temporal error is negligible. Since the two schemes also use different spatial approximation orders, these figures are separate stability tests rather than a controlled comparison of temporal accuracy. The computations provide numerical evidence of the discrete dissipation mechanism at the tested step sizes, not a proof for arbitrary step sizes.

Table 4: Second-order unforced energy test: $K = 8$, $T = 0.5$.

Δt	$E_h(T)/E_h(0)$	$\max_n \epsilon_{\text{bal}}^n$
1/16	1.1117×10^{-4}	3.71×10^{-11}
1/32	2.5984×10^{-5}	7.77×10^{-13}
1/64	1.0438×10^{-5}	2.67×10^{-13}

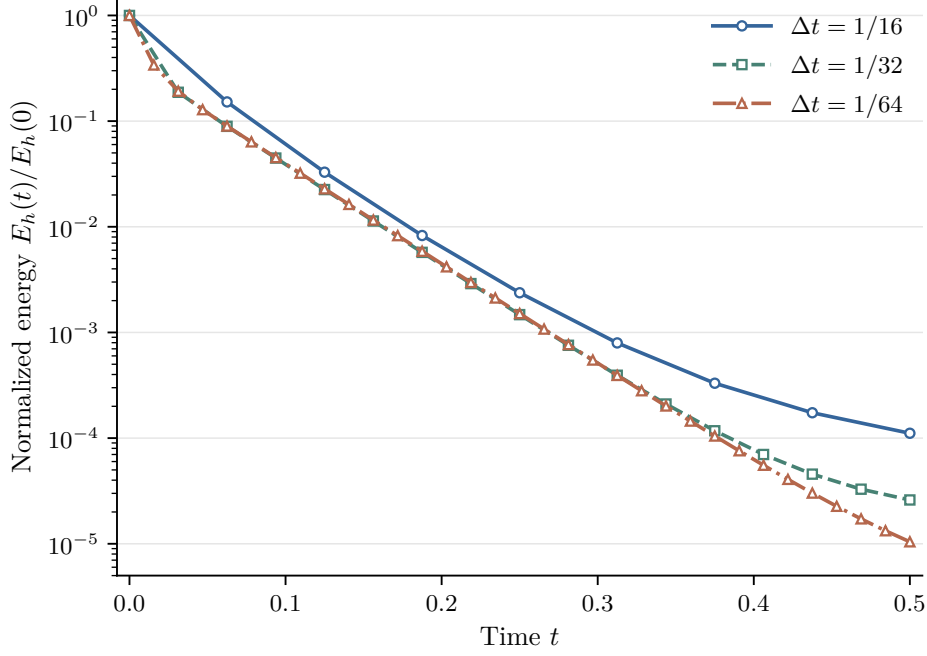


Figure 4: Normalized discrete energy for the second-order scheme in the unforced relaxation test, $K = 8$ and $T = 0.5$. The logarithmic vertical scale matches Figure 3. Symbols represent all computed time levels without smoothing. The same discrete initial state is used for all three time steps.

6.4 Magnetically forced flow in a straight channel

We next apply the second-order scheme and the five-block Algorithm 3 to three-dimensional ferrofluid transport with prescribed inflow. The configuration is motivated by the channel-pumping examples of Dong et al. [2], while retaining the magnetic diffusion, compatible finite element spaces, and midpoint/extrapolation scheme of the present work. The parameters and driving waveform below define our computational example; they are not a reproduction of all the parameters in that reference. All quantities are dimensionless.

Domain, material parameters, and initial data. The straight channel is

$$\Omega_c = (0, 1) \times (0, 0.2) \times (0, 0.2).$$

We denote the inlet $x = 0$, outlet $x = 1$, and remaining solid walls by Γ_i , Γ_o , and Γ_w , respectively. The material parameters, also used in the stepped channel below, are

$$\begin{aligned} \rho = \mu_0 = \chi_0 = 1, \quad \kappa = \zeta = 0.01, \quad \eta = 0.02, \\ \eta' = \lambda' = 0.001, \quad \tau = 0.1, \quad \sigma = 0.005. \end{aligned} \tag{4}$$

The body sources are $\mathbf{f} = \mathbf{g} = \mathbf{r} = \mathbf{0}$. The initial velocity, angular velocity, magnetization, and magnetic potential are zero, as are the pressure and auxiliary variables. These data are compatible with the initially vanishing inlet and magnetic forcing.

Inlet and magnetic forcing. Introduce the smooth ramp

$$R(s) = \begin{cases} 0, & s \leq 0, \\ 10s^3 - 15s^4 + 6s^5, & 0 < s < 1, \\ 1, & s \geq 1. \end{cases} \tag{5}$$

The inlet profile is

$$\mathbf{u}_D = (0.25R(t/0.4) 16Y(1-Y)Z(1-Z), 0, 0)^\top, \quad Y = \frac{y}{0.2}, \quad Z = \frac{z}{0.2}. \quad (6)$$

It vanishes at the inlet edges and is imposed by nodal interpolation and finite element Dirichlet lifting.

Eight dipoles are placed outside the channel at

$$\mathbf{x}_i^- = (0.2i, -0.15, 0.1), \quad \mathbf{x}_i^+ = (0.2i, 0.35, 0.1), \quad i = 1, \dots, 4,$$

with directions $\mathbf{d}_i^- = (0, 1, 0)^\top$ and $\mathbf{d}_i^+ = (0, -1, 0)^\top$. We prescribe

$$\Phi_a(\mathbf{x}, t) = \sum_{i=1}^4 a_i(t) \sum_{\epsilon \in \{-, +\}} \frac{\mathbf{d}_i^\epsilon \cdot (\mathbf{x}_i^\epsilon - \mathbf{x})}{|\mathbf{x}_i^\epsilon - \mathbf{x}|^3}, \quad \mathbf{h}_a = \nabla \Phi_a, \quad (7)$$

$$a_i(t) = C_a A R \left(\frac{t - 0.2}{0.4} \right) \left[0.7 + 0.3 \cos \left(2\pi \left(t - \frac{0.2i}{0.8} \right) \right) \right], \quad C_a = 3 \times 10^{-4}. \quad (8)$$

The dipole positions are fixed; the phase shifts act on their time-dependent strengths. The magnetic forcing has period one after the ramp. We take $A = 1$ for the magnetic case and $A = 0$ for a zero-field control with the same inlet data. In the notation of the model, $\mathbf{H}_e = -\mu_0 \mathbf{h}_a$. Since all sources lie outside the fluid, Φ_a is harmonic there. Consequently, the magnetic-potential load is

$$(\mathbf{G}_e, \nabla \psi_h), \quad \mathbf{G}_e = \mu_0 \partial_t \mathbf{h}_a + \frac{\mu_0 + \chi_0}{\tau} \mathbf{h}_a.$$

Open-boundary formulation. We impose $\mathbf{u} = \mathbf{0}$ on Γ_w and the gradient-traction condition

$$((\eta + \zeta) \nabla \mathbf{u} - p \mathbf{I}) \mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_o, \quad (9)$$

where p is the physical pressure and the computed pressure slot is $\tilde{p} = p - \mu_0 \mathbf{m} \cdot \mathbf{H}/2$. The outlet fixes the pressure level, so no pressure-mean constraint is added in these open-channel runs. On the entire boundary we prescribe

$$\begin{aligned} \boldsymbol{\omega} &= \mathbf{0}, \quad \mathbf{m} \cdot \mathbf{n} = 0, \quad \mathbf{k} \times \mathbf{n} = \mathbf{0}, \\ (\mathbf{H} + \mathbf{m}) \cdot \mathbf{n} &= \mathbf{h}_a \cdot \mathbf{n}, \quad \int_{\Omega_c} \varphi_h \, dx = 0. \end{aligned} \quad (10)$$

The tangential trace of $\mathbf{z}_h$ is zero on Γ_w and free on the inlet and outlet. The trace of ϖ_h is zero on $\Gamma_i \cup \Gamma_w$ and free on Γ_o . The weak definition of ϑ_h includes the prescribed normal load:

$$(\vartheta_h, v_h) = (\nabla \varphi_h, \nabla v_h) - \langle \mathbf{h}_a \cdot \mathbf{n}, v_h \rangle_{\partial \Omega_c}.$$

The nine volume equations and their five-block coupling are retained. The open-boundary extension includes the term $\frac{\mu_0}{2} (\text{curl } \mathbf{z}_h, \nabla \psi_h)$ in the magnetic-potential equation. To implement (9) with the gradient/curl form of the momentum equation, its residual also contains

$$\begin{aligned} \mathcal{R}_{\Gamma_o}(\mathbf{v}_h) &= -\langle \mathbf{t}_\Delta, \mathbf{v}_h \rangle_{\Gamma_o} + \frac{\rho}{2} \langle (\mathbf{u}_* \cdot \mathbf{n}) \mathbf{u}, \mathbf{v}_h \rangle_{\Gamma_o}, \\ \mathbf{t}_\Delta &= -\zeta (\nabla \mathbf{u})^\top \mathbf{n} + 2\zeta \mathbf{n} \times \boldsymbol{\omega} \\ &\quad + \frac{\mu_0}{2} [(\mathbf{m}_* \cdot \mathbf{H}) \mathbf{n} + \mathbf{n} \times (\mathbf{H} \times \mathbf{m}_*)]. \end{aligned} \quad (11)$$

Here the unstarred fields in the evolution residuals are midpoint values, while stars denote the prescribed nonlinear-startup or extrapolated coefficients. The inlet lifting and the normal load in the ϑ_h relation use endpoint data. These boundary conditions extend the homogeneous setting of the analysis. The closed-boundary energy estimates are not asserted for this driven open-boundary problem.

Discretization and results. We use the second-order finite element spaces of Section 6.1, $\Delta t = 0.005$, and $T = 2$. The initial step solves the nonlinear midpoint equations; subsequent steps use the specified extrapolation and Algorithm 3, including the fifth-block feedback. Assembly uses quadrature degree eight. The Schur solves use relative and absolute tolerances 10^{-11} and 10^{-16} ; the nonlinear startup uses relative tolerance 10^{-9} with an absolute residual floor of 10^{-14} . The channel meshes contain $30K^3$ tetrahedra, obtained from a $5K \times K \times K$ Cartesian partition. The magnetic runs use $K = 4, 8, 12$, and the control uses $K = 12$.

Figures 5 to 7 show the $K = 12$ magnetic solution at $t = 2$. The velocity retains a predominantly axial profile, whereas the magnetization has a nonuniform distribution set by the dipole field and its coupled response. The particle angular velocity is appreciable in the sheared portions of the flow; it is distinct from the fluid vorticity $\text{curl } \mathbf{u}_h$.

At $T = 2$, native volume- L^2 differences between the $K = 8$ and $K = 12$ solutions, normalized by the $K = 12$ norm, are

Field	u	m	ω	H
Relative difference (%)	0.13524	3.29718	0.71893	3.14611

These are adjacent-mesh differences, not errors against an exact solution. They were integrated on a common refinement and checked with quadrature degrees four and six. The same-mesh zero-field control has zero magnetization and magnetic field. Its differences from the magnetic run are 0.00725% in velocity and 0.15803% in angular velocity in the same native norm, with the magnetic solution as denominator. These small increments have not been independently established to be mesh-converged. The prescribed inlet fixes the through-flow; this comparison concerns field response under magnetic forcing and does not establish a net magnetic pumping gain.

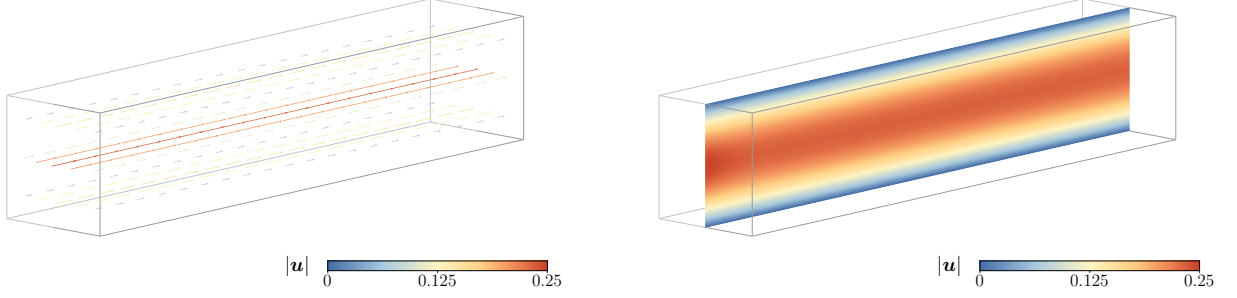


Figure 5: Straight-channel flow at $t = 2$ on $K = 12$: velocity vectors (left) and the velocity magnitude on $z = 0.1$ (right).

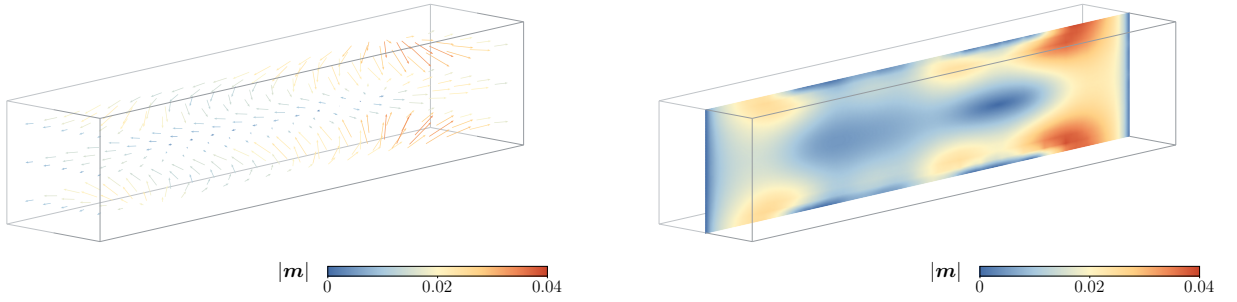


Figure 6: Straight-channel flow at $t = 2$ on $K = 12$: magnetization vectors (left) and the magnetization magnitude on $z = 0.1$ (right).

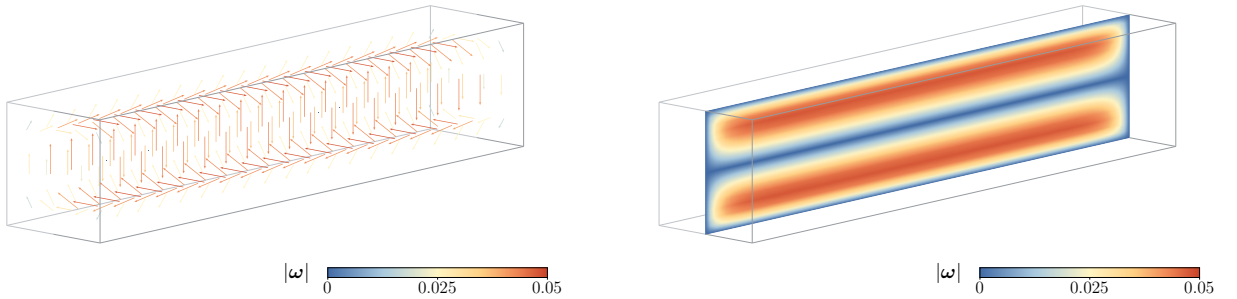


Figure 7: Straight-channel flow at $t = 2$ on $K = 12$: particle angular velocity vectors (left) and their magnitude on $z = 0.1$ (right). The arrows display instantaneous vectors, not particle trajectories.

6.5 Magnetically forced flow over a backward-facing step

To examine the same coupled solver in a channel with a sudden expansion, we consider

$$\Omega_s = (0, 1) \times (0, 0.2) \times (0, 0.2) \setminus ([0, 0.2] \times [0, 0.2] \times [0, 0.1]). \quad (12)$$

The upstream bottom is raised to $z = 0.1$ for $x \leq 0.2$. The fluid volume is 0.036; the inlet $x = 0$, $z > 0.1$ has area 0.02, and the outlet $x = 1$ has area 0.04. The vertical step face and all remaining boundary faces belong to the no-slip wall Γ_w . The geometry and the dipole locations follow the three-dimensional step configuration in [2].

The material parameters, zero initial data, zero body sources, outlet traction, and magnetic and auxiliary boundary conditions are those stated in Section 6.4, with Ω_c replaced by Ω_s . The inlet profile is

$$\mathbf{u}_D = (0.25R(t/0.4)16Y(1-Y)Z_s(1-Z_s), 0, 0)^\top, \quad Y = \frac{y}{0.2}, \quad Z_s = \frac{z-0.1}{0.1}. \quad (13)$$

It is supported on the upper half of the inlet and vanishes on its edges. The inlet and homogeneous wall values are imposed separately, so the polynomial inlet extension is not applied to the vertical step face. After ramp-up, the analytic inlet flux is $1/450$, one half of the straight-channel value because the inlet area is halved.

The applied field retains (7)–(8), with $A = 1$, but uses

$$\mathbf{x}_i^- = (0.2i, -0.2, 0.1), \quad \mathbf{x}_i^+ = (0.2i, 0.4, 0.1), \quad i = 1, \dots, 4, \quad C_a = 7 \times 10^{-4}. \quad (14)$$

The directions remain $\mathbf{d}_i^\pm = \mp(0, 1, 0)^\top$, with the upper sign corresponding to the upper row. Thus the waveform, ramp times, period, and phase speed are unchanged, while the magnetic-source distance and prefactor are explicitly different. In particular, the relaxation time remains $\tau = 0.1$ and the diffusion coefficient remains $\sigma = 0.005$.

Mesh and time stepping. We use the same second-order spaces, nonlinear midpoint startup, subsequent extrapolation, quadrature, and solver tolerances as in the straight channel. The time step is 0.005 and the simulation runs to $T = 2$. A conforming tetrahedral mesh is formed from a $5K \times K \times K$ background partition, with the solid cells removed. Before extracting the fluid mesh, its logical axial coordinate ξ is mapped to x by the continuous piecewise affine function specified by

ξ	0	0.1	0.2	0.6	1
x	0	0.15	0.2	0.4	1.

This grades the mesh near the step; the transverse coordinates remain uniform. The resulting mesh has $27K^3$ tetrahedra. The following visualizations use $K = 4$, namely 1728 tetrahedra.

Field visualization and interpretation. Figures 8 to 10 use the actual saved time layer $t = 0.715$ (step 143), rather than a temporally interpolated state. All three fields use this same time. The velocity spreads into the larger downstream cross-section after the step. The magnetization exhibits spatially varying direction and magnitude, while the angular-velocity distribution is influenced by the shear near the step and the channel walls. The displayed angular velocity is the particle spin.

The right panels show the full three-component magnitude on $y = 0.1$. The magnetization arrows are sampled on that same plane; the velocity and angular-velocity arrows use sparse three-dimensional locations. Solid-region samples are excluded. The linear arrow scaling and the color scales are fixed over the stored time interval. Point values come from the original finite element fields without RT vertex averaging; interpolation in the filled color surface is used only for display, not for evaluating finite element norms. The color limits for $\mathbf{u}$, $\mathbf{m}$, and $\boldsymbol{\omega}$ are 0.25, 0.04, and 0.04, respectively.

The 400-step $K = 4$ trajectory passed the stated algebraic and boundary checks: its maximum recomputed relative residual was 2.03×10^{-10} , and its largest recorded relative inlet–outlet flux imbalance was 1.12×10^{-14} . A full-interval $K = 4$ comparison with $\Delta t = 0.0025$ gave endpoint relative native volume- L^2 differences below 2.9×10^{-5} for $\mathbf{u}$, $\mathbf{m}$, $\boldsymbol{\omega}$, and $\mathbf{H}$. These checks assess the implementation and time-step sensitivity. At the displayed spatial resolution, the figures are a qualitative illustration; quantitative mesh independence of the step-flow observables remains to be assessed. In particular, no resolved reattachment length, corner extremum, or magnetic pumping efficiency is inferred from these plots. The nonconvex geometry and open boundary conditions also lie outside the homogeneous convex-domain setting used in the analysis.

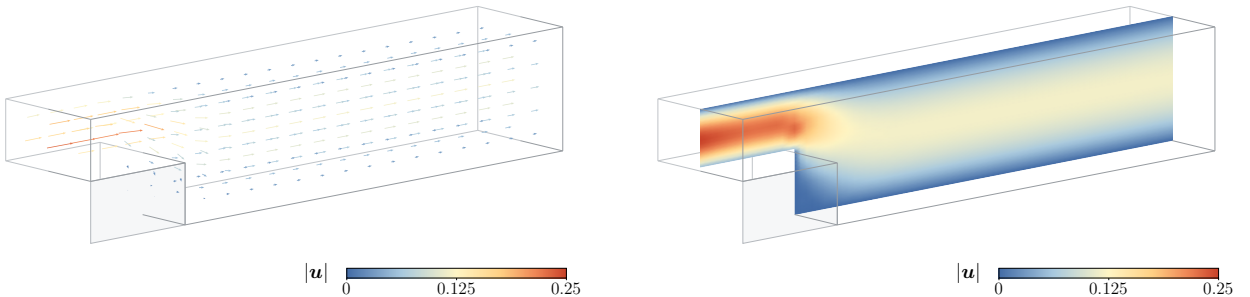


Figure 8: Backward-facing-step flow at $t = 0.715$ on $K = 4$: velocity vectors (left) and the velocity magnitude on $y = 0.1$ (right). The unfilled upstream region is the solid step.

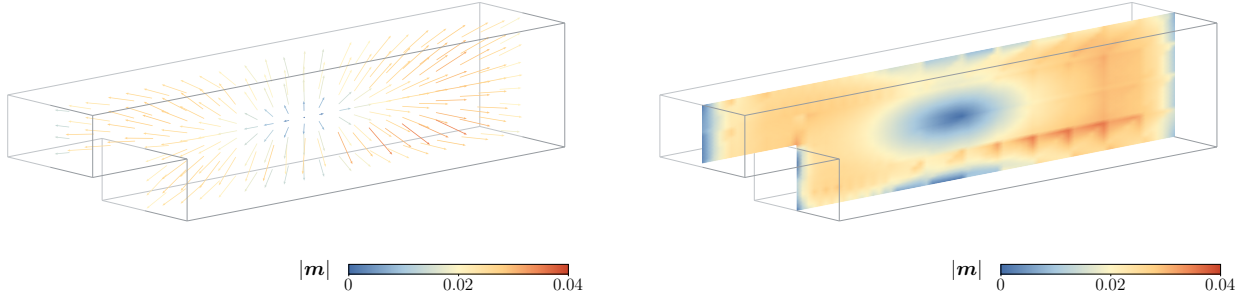


Figure 9: Backward-facing-step flow at $t = 0.715$ on $K = 4$: magnetization vectors (left) and their full magnitude (right), both sampled on $y = 0.1$.

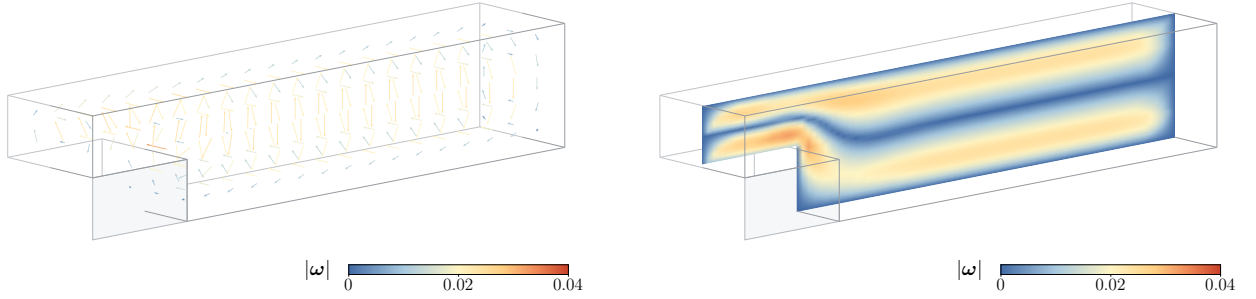


Figure 10: Backward-facing-step flow at $t = 0.715$ on $K = 4$: particle angular velocity vectors (left) and their magnitude on $y = 0.1$ (right).

References

- [1] I. A. Baratta, J. P. Dean, J. S. Dokken, M. Habera, J. S. Hale, C. N. Richardson, M. E. Rognes, M. W. Scroggs, N. Sime, and G. N. Wells, DOLFINx: The next generation FEniCS problem solving environment, 2023.
- [2] X. Dong, H. Huang, Y. Huang, and X. Yang, A fully discrete finite element method with second-order temporal accuracy for the Rosensweig model, *J. Comput. Phys.*, 549 (2026), 114617.